Math 306 Group Quiz 5

Names: ____

1. Let U be a subspace of a finite dimensional inner product space V. The orthogonal complement is

$$U^{\perp} = \{v \in V : \langle u, v \rangle = 0 \text{ for all } u \in U\}.$$

a. Show that $U^{\perp}$ is a subspace.

b. Using the standard inner product on $\mathbb{C}^3$, find a basis for $\operatorname{span}\{(1, 1, -1)\}^{\perp}$.

c. Suppose $v_1, \ldots, v_k$ is an orthonormal basis for U and $v_1, \ldots, v_k, v_{k+1}, \ldots, v_n$ is an orthonormal basis for V. Prove that $v_{k+1}, \ldots, v_n$ is a basis for $U^{\perp}$.

d. Prove that $\left(U^{\perp}\right)^{\perp} = U$.